Demography & Health Statistics

Public Health & Epidemiology of Infectious Diseases

Lecture 2

Learning Objectives

Core Concepts

- Define health statistics and demography.
- Identify key data sources: vital, morbidity, service, census.
- Explain core indicators (CBR, ASFR, TFR).

Applied Analysis

- Describe Palestine’s demographic profile.
- Analyze population pyramids
- **Link population characteristics to infectious disease risk and IPC planning.**

What is Demography?



Size

The total number of persons in a given area.



Structure

Composition by age, sex, and other variables.



Distribution

Where people live (urban vs. rural, density).

The Purpose of Data

"Information for Health"

The health status of a population is assessed by collecting data on important events: births, deaths (mortality), and sickness (morbidity).

The Data Gap Challenge

Obstacle to Management

The lack of reliable data in developing countries is a major barrier to effective health care management.

Without data, decision-makers cannot:

- Plan services accurately.
- Implement targeted interventions.
- Evaluate program success.

Public health policy planning



Source 1: Vital Statistics

Medical records and vital statistics



Civil Registration

Vital statistics are records of vital events obtained by continuous registration systems.

Key Events:

- Births (Live births)
- Deaths (Mortality)
- Marriages
- Divorces

Use: Generating birth and mortality rates for the population.

Source 2: Morbidity Statistics



Occurrence

Data on the frequency of sickness in a community.



Severity

Data on the impact and intensity of diseases.



Sources

Hospitals, clinics, private practitioners, notification systems.

Note: Morbidity data is harder to collect and interpret than vital statistics.

Source 3 & 4: Service & Census

3. Health Service Statistics

Derived from the daily operations of the health system.

- Hospital admissions/discharges
- Outpatient visits
- Vaccination coverage records
- Resource utilization

4. The Census

The essential population base for calculating rates.

- Total count of population
- Age and sex distribution
- Personal data (education, employment)
- Usually conducted every 10 years

Source 5: Other Sectors

Health is determined by more than just healthcare. We need data from:



Education

Literacy rates, especially female literacy.



Public Works

Housing, water supply, sanitation access.



Agriculture

Food production, distribution, food security.



Economy

Poverty rates, employment, economic development.

The Census: The Denominator

The Foundation of Epidemiology

To calculate any "Rate" (Birth Rate, Death Rate, Disease Rate), we need a denominator.

The Census provides the **Population at Risk**.

- Held periodically (e.g., every 10 years).
- Provides the "Mid-Year Population" estimate.
- Critical for planning health services and schools.



Fertility Metrics

Measuring Population Growth

Crude Birth Rate (CBR)

Definition

The number of live births per 1,000 population in a given year.

$$\text{CBR} = \frac{\text{Live Births}}{\text{Mid-Year Population}} \times 1000$$

Why "Crude"? It includes men, children, and the elderly in the denominator



Calculation Example

Scenario: A community in 1960.

18,247
Total Births

/

985,210
Population

=

18.5
per 1,000

Interpretation: There were 18.5 births for every 1,000 people in that year.

Age-Specific Fertility Rate (ASFR)

Refining the Measure

ASFR measures the number of live births per 1,000 women in a **specific age group**.
The likelihood of having a child varies significantly by age (e.g., 20-24 vs 45-49).
Standard Age Groups: Seven 5-year groups covering the reproductive years (15-49).

$$\text{ASFR} = \frac{\text{Births to women age X}}{\text{Total women age X}} \times 1000$$

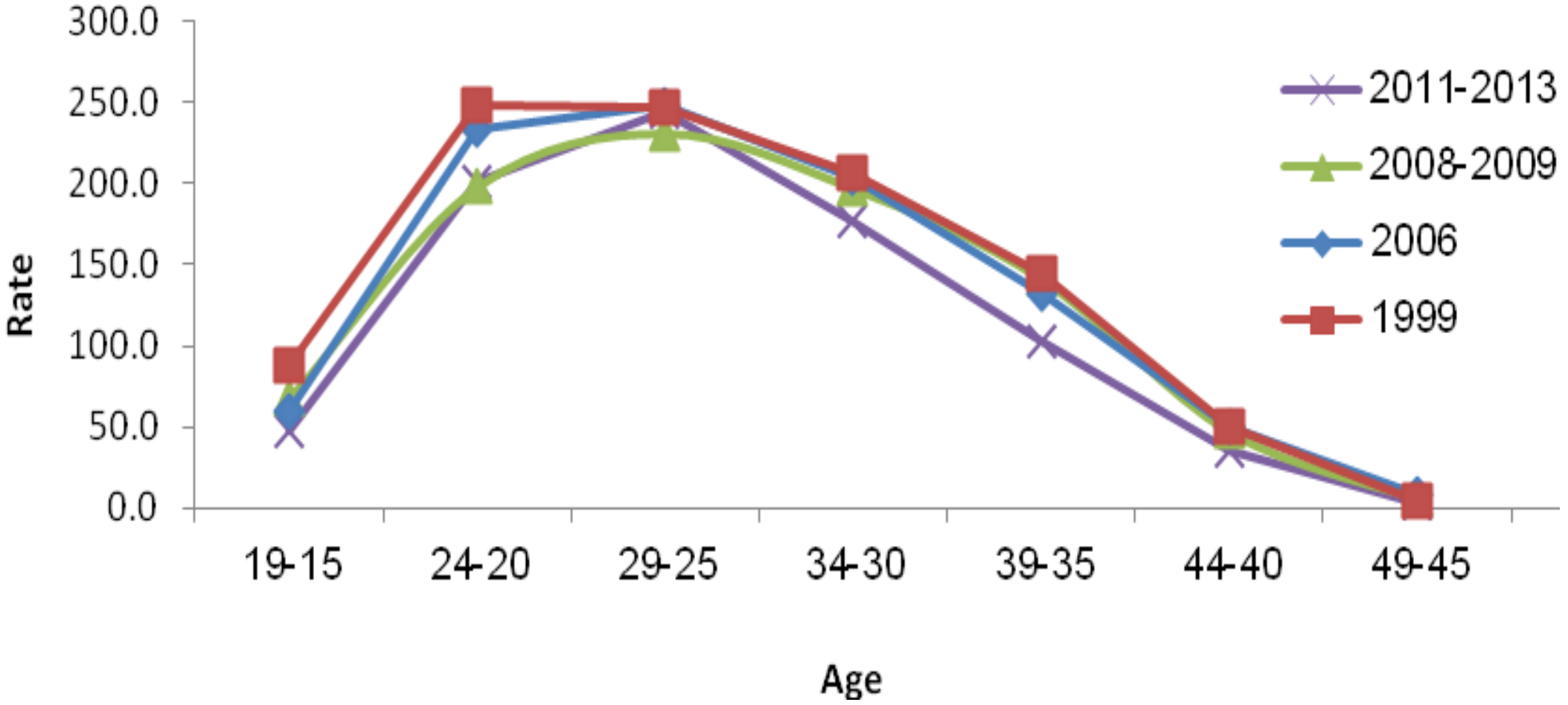
ASFR Calculation Table

Example: Women aged 20-24 in Country X (2011)

Age Group	Live Births (Numerator)	Women Population (Denominator)	Calculation	ASFR (per 1000)
15-19	...	...	...	...
20-24	**72,000**	**310,000**	**(72k / 310k) * 1000**	**232.3**
25-29	...	...	...	...

Interpretation: In this country, 232 out of every 1,000 women aged 20-24 gave birth in 2011.

There has been a decline in the specific fertility rate, particularly in the early reproductive years (15-24 years).



Age-Specific Fertility Rates in State of Palestine, Selected Years

Total Fertility Rate (TFR)



The Gold Standard Indicator

The TFR represents the average number of children a woman would have by the end of her reproductive life if she experienced the current age-specific fertility rates.

Significance: It tells us about family size and population growth potential.

Replacement Level = 2.1 children per woman.

Calculating TFR

$$TFR = 5 \times \sum \left(ASFR_{15-49} \right) / 1000$$

Steps:

1. Calculate ASFR for each of the 5-year age groups.
2. Sum all the ASFRs together.
3. Multiply by 5 (since each group covers 5 years).
4. Divide by 1000 (to get rate per woman, not per 1000 women).

Example:

If sum of ASFRs = 657 per 1000 women.

TFR = 5 * 657 = 3285.

Divide by 1000 = **3.28 children per woman**.

Factors Affecting Fertility



Age at Marriage

Younger marriage = longer reproductive window.



Education

Higher female education correlates with lower TFR.



Socioeconomic

Economic development often reduces family size.



Spacing

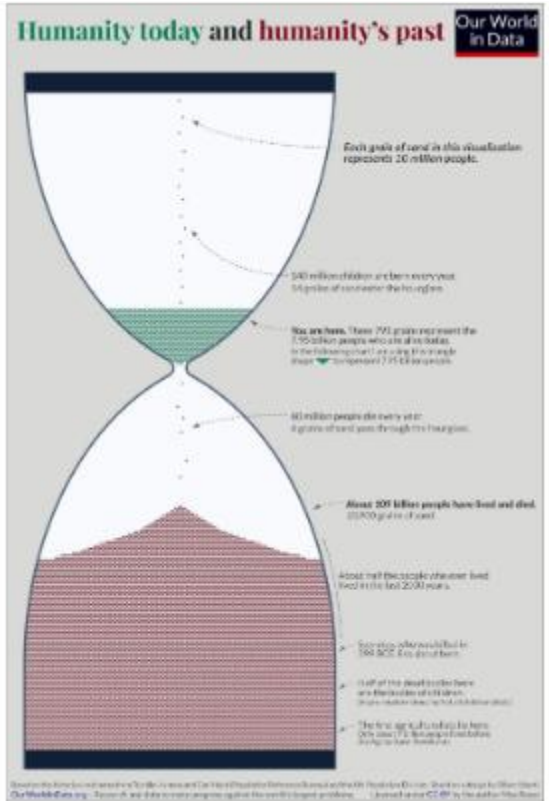
Access to family planning and birth spacing.

Mortality Measures

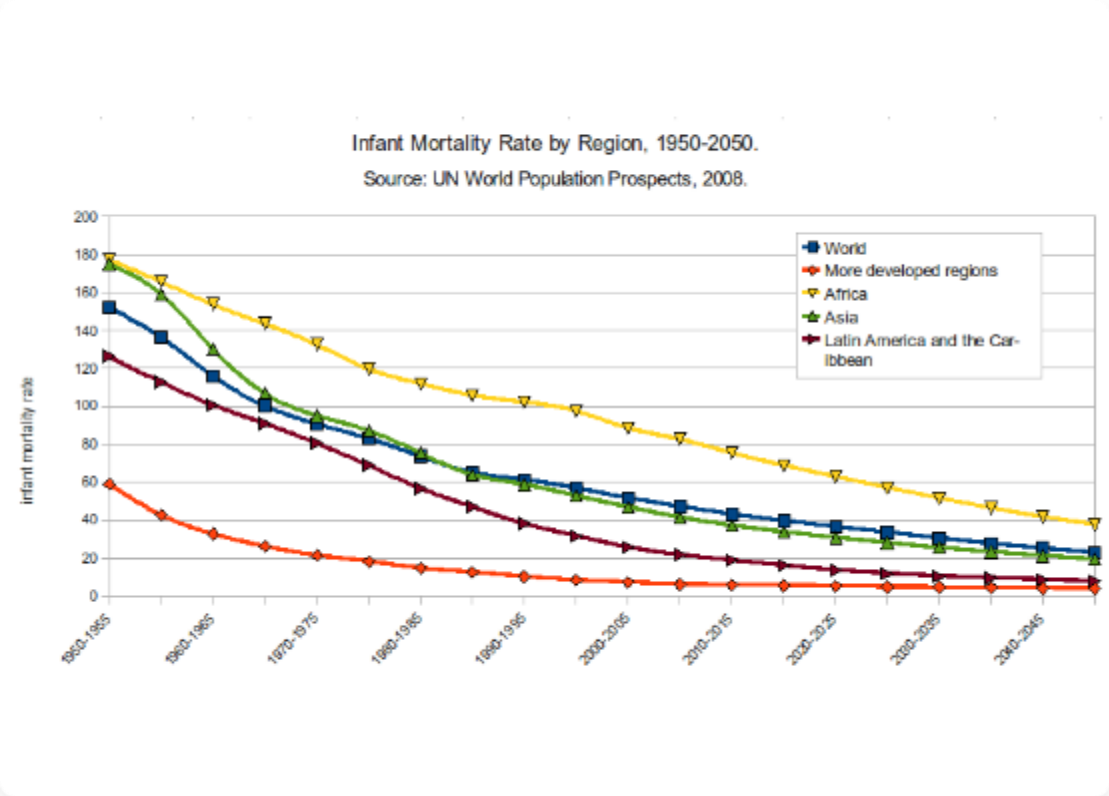
Crude Death Rate (CDR)

Number of deaths per 1,000 population in a given year.

Like CBR, it is "crude" because it doesn't account for the age structure (a retirement community will have a high CDR naturally).



Infant Mortality Rate (IMR)



A Sensitive Indicator

IMR is the number of deaths of infants (under 1 year) per 1,000 **live births**.

It is widely used as an indicator of a country's overall health and development status.

Causes of Decline: Vaccines, sanitation, MCH services, nutrition.

Population Theory

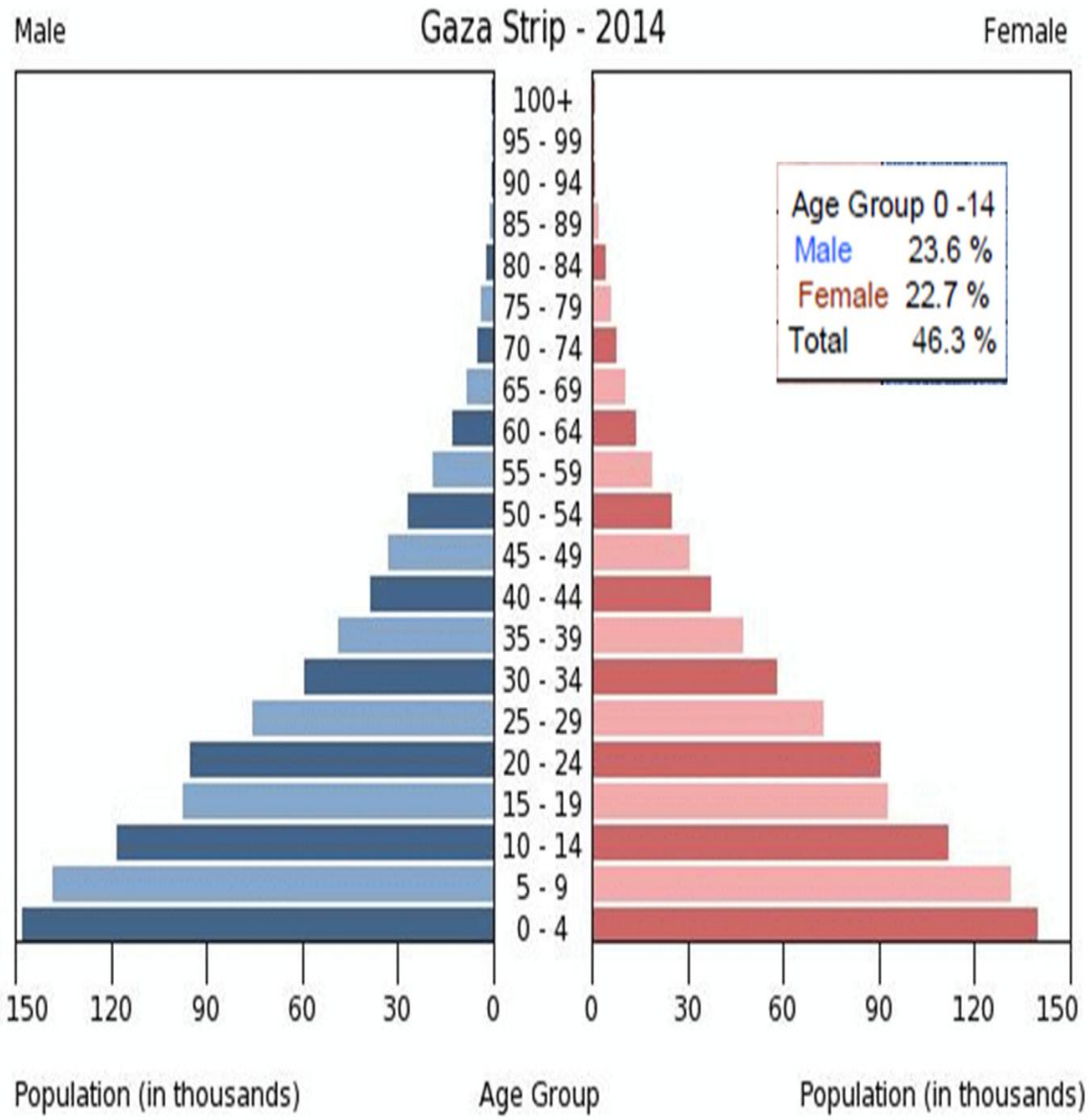
Pyramids & The Demographic Transition

The Population Pyramid

Graphical Representation

A histogram showing the percentage distribution of a population by age and sex.

- **Males:** Usually on the left.
- **Females:** Usually on the right.
- **X-axis:** Percentage or population count.
- **Y-axis:** Age groups (usually 5-year intervals).



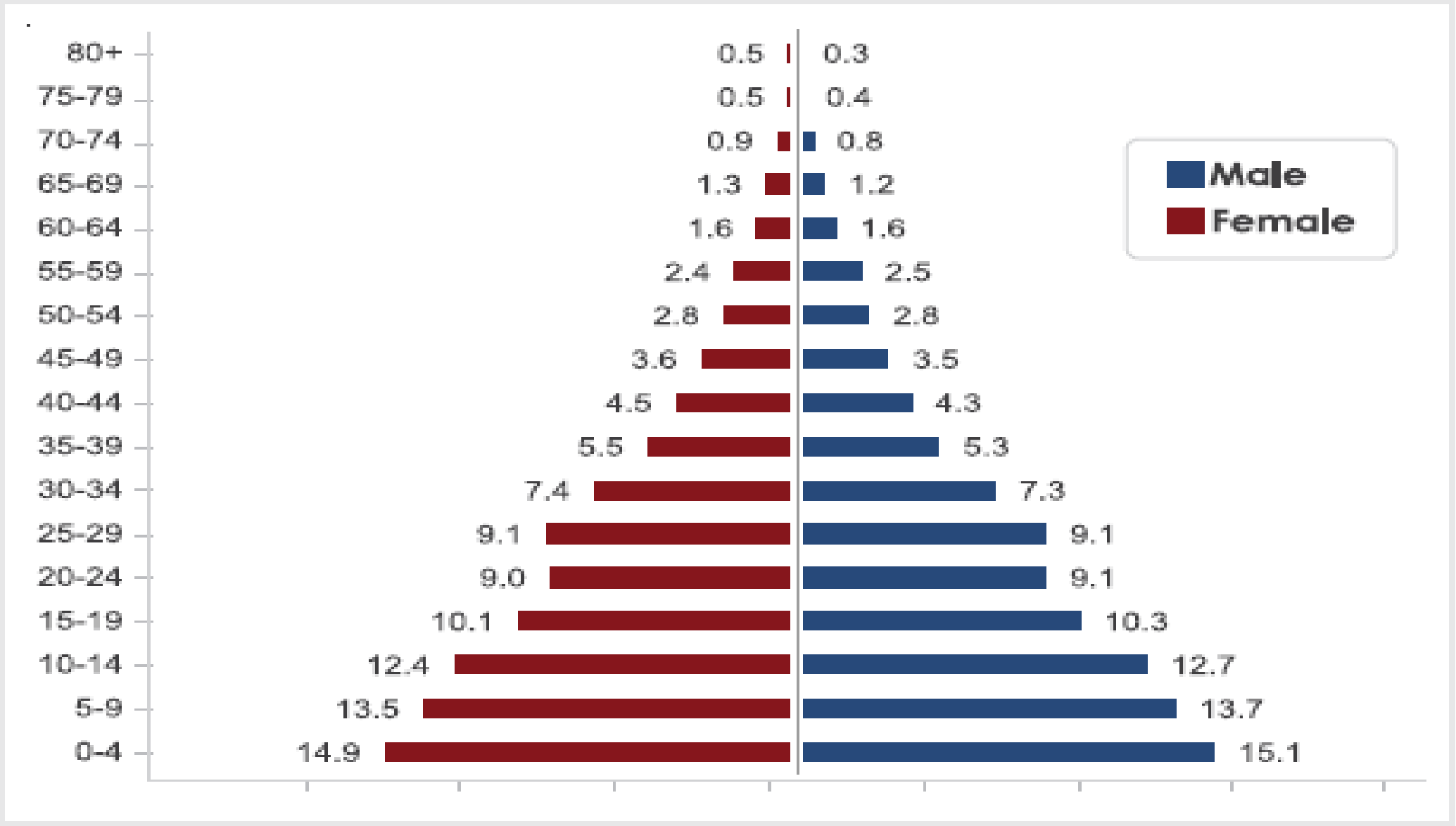
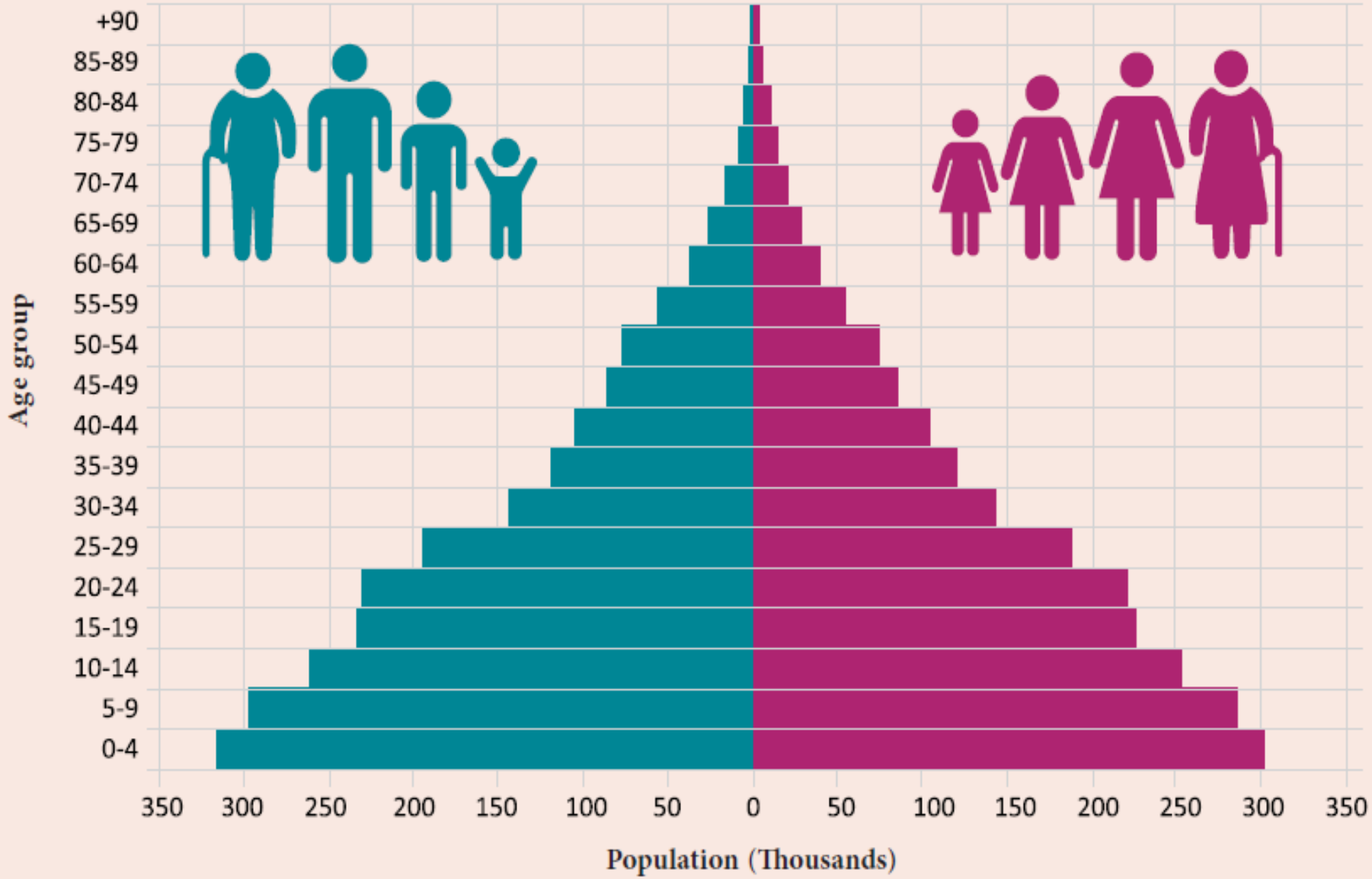


Figure (1) : Population Pyramid of Gaza Strip by Gender & Age Group 2020

Population pyramid



Interpreting the Base

Wide Base

= High Birth Rate

A large percentage of the population is under age 15.

Implication: High dependency ratio, need for schools and pediatric health services.

Narrow Base

= Low Birth Rate

Fewer children are being born.

Implication: Shrinking workforce in the future.

Interpreting the Apex

Tapering Apex

Rapid narrowing at older ages indicates high mortality and shorter life expectancy.

Typical of developing countries.

Broad Apex

Indicates a "graying" population with high life expectancy.

Requires focus on chronic disease management and geriatric care.



Interpreting Shapes



Expansive

Broad base, narrow top. High birth rate, high death rate, short life expectancy. (e.g., Palestine, Niger)



Stationary

Rectangular shape. Declining birth rate, low death rate, long life expectancy. (e.g., USA)



Constrictive

Narrow base. Very low birth rate, shrinking population. (e.g., Japan, Germany)

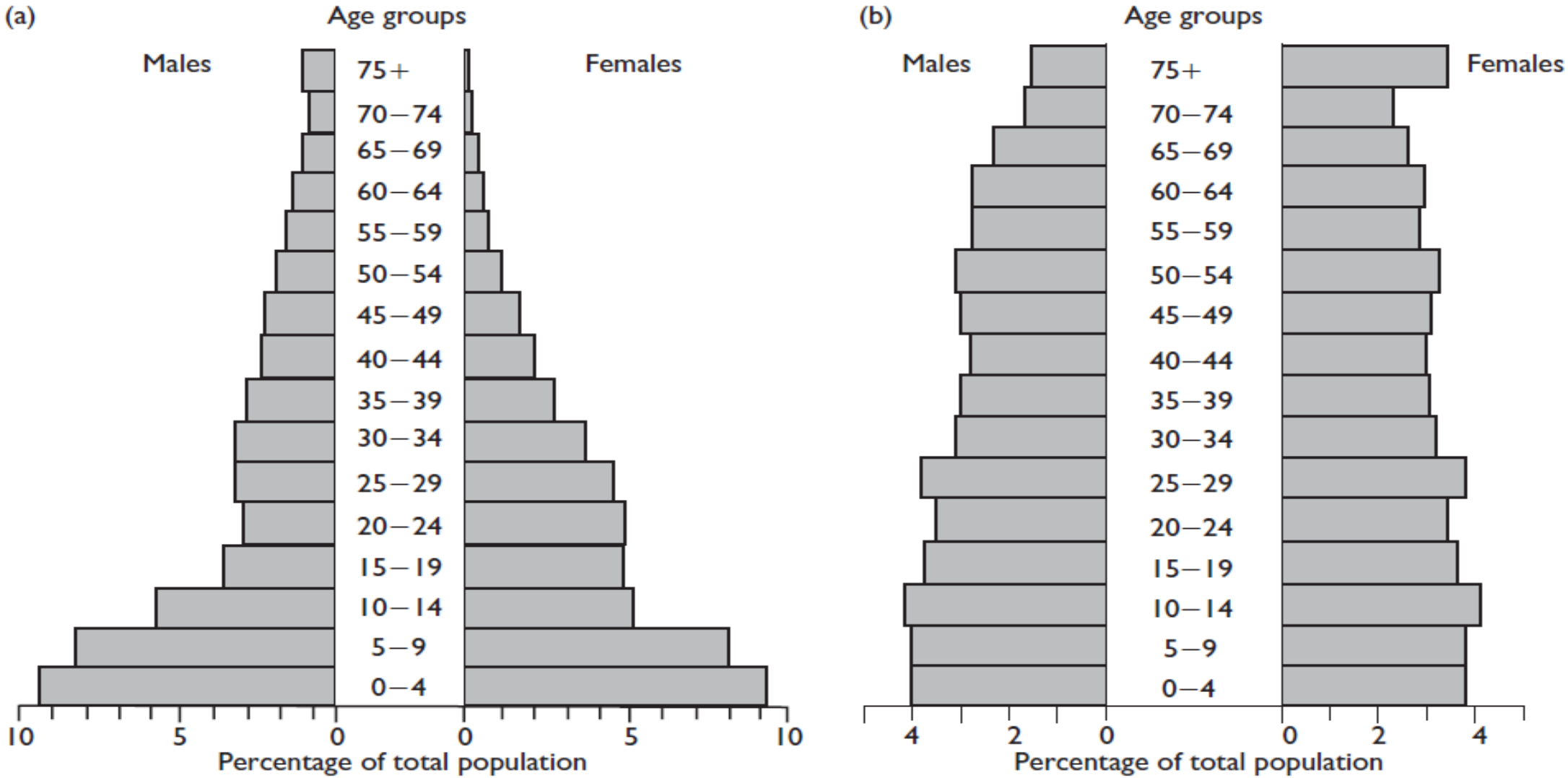
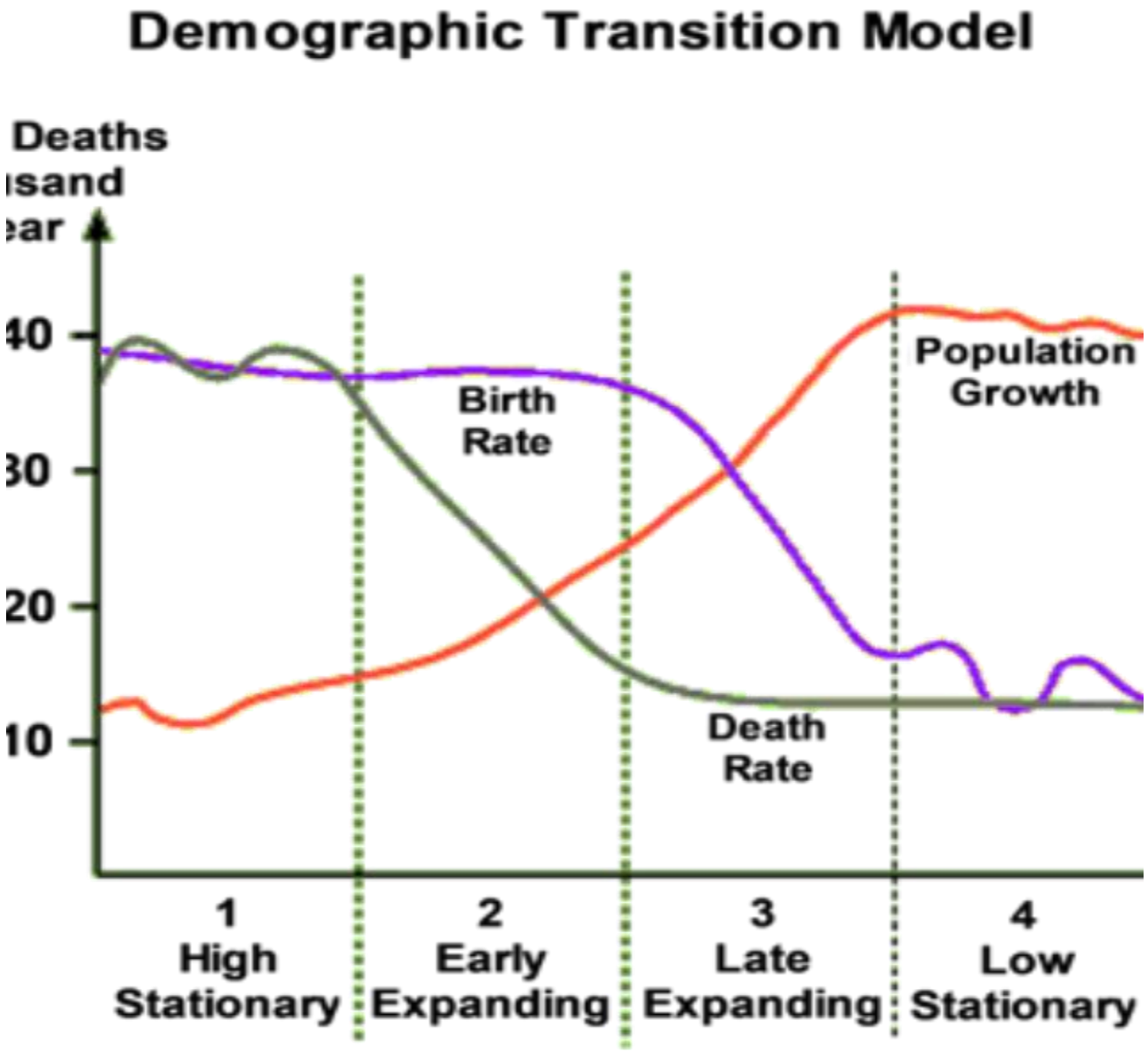


Figure 3 : Population pyramids showing the distribution of males and females amongst different age groups (a) in a developing country, (b) in an industrialized country.

Demographic Transition Model

A model describing the transition from high birth/death rates to low birth/death rates as a country develops.

Based on historical trends in industrialized societies (observed by Warren Thompson, 1929).



Stage 1: High Stationary

High Birth Rate

Lack of family planning, religious beliefs, need for farm labor.

High Death Rate

Disease, famine, lack of clean water, war.

Result

Population growth is slow and fluctuating.

Example: Pre-industrial societies, isolated tribes.

Stage 2: Early Expanding

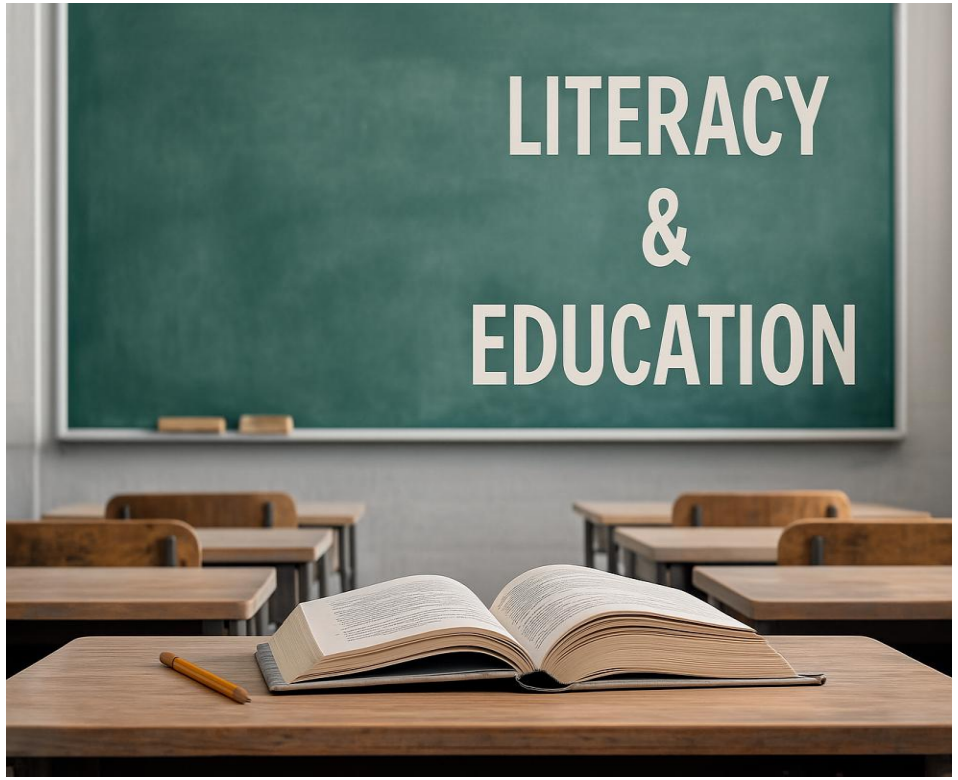
The "Explosion" Begins

- **Death Rate:** Falls rapidly.
- **Birth Rate:** Remains high.
- **Population:** Rises steadily.

Why? Improved hygiene, sanitation (boiling water), transport of food, vaccines.



Stage 3: Late Expanding



Growth Slows Down

- **Death Rate:** Continues to fall.
- **Birth Rate:** Starts to fall.
- **Population:** Rising, but slower.

Why? Family planning, changing status of women, reduced infant mortality (less need for "replacement" births).

Stage 4: Low Stationary

Characteristics

- Low Birth Rate
- Low Death Rate
- Steady Population

Examples

USA, Sweden, Japan, Britain.

Focus shifts to chronic diseases and lifestyle factors.

Stage 5: Declining?

Declining Population

Birth rate drops below death rate. Death rate rises slightly due to aging population.

Examples: Germany, Japan

The Palestinian Profile

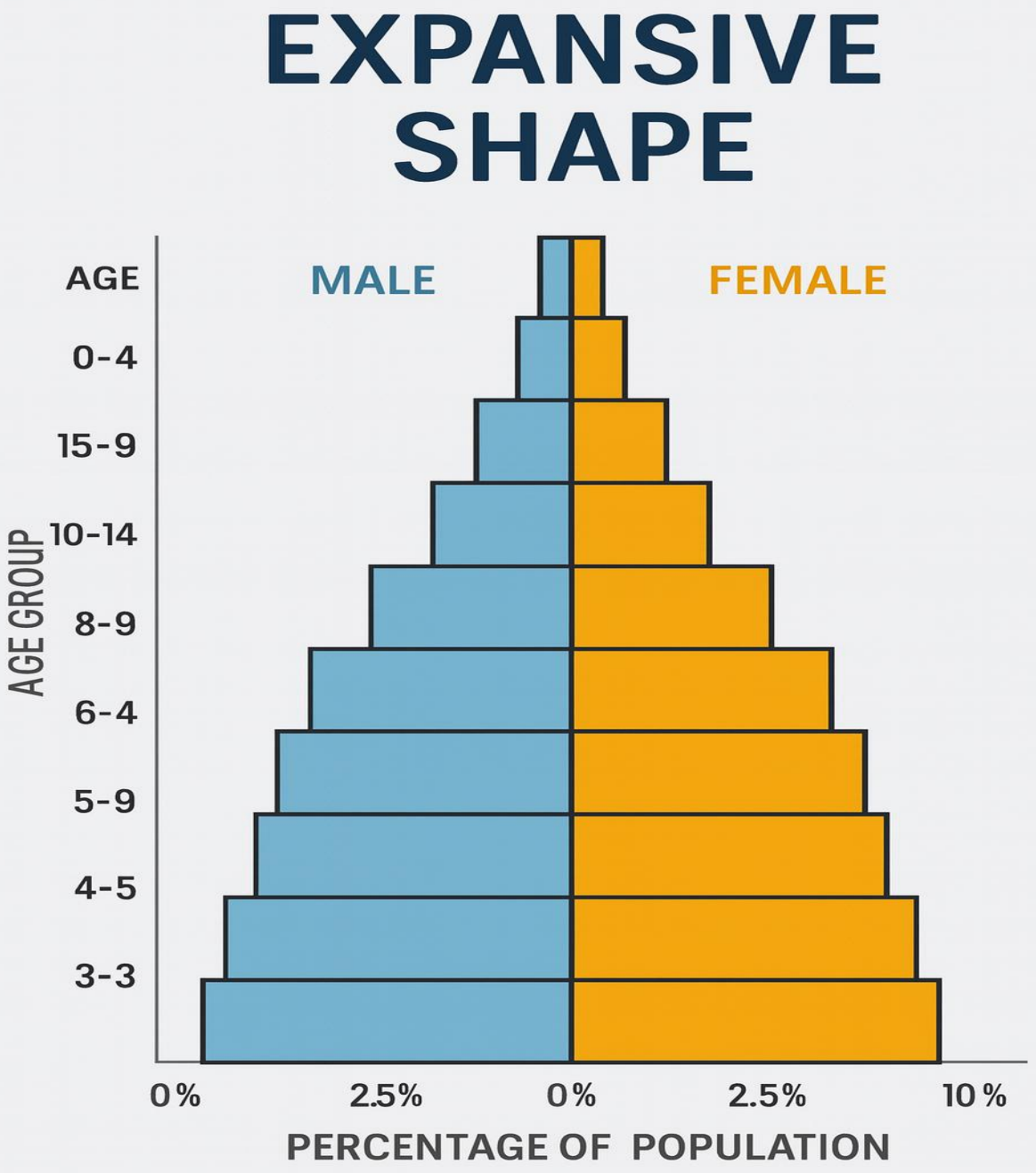
Local Demographics & Health

Palestine's Age Structure

Classic "Expansive" Shape

- **Broad Base:** Large population of children (high birth rate).
- **Tapering Top:** Small elderly population.

This structure indicates a young, growing population with specific health needs (MCH, Schools).



The "Youth Bulge"

Median age divides the population into two equal halves.

18
Gaza Median Age

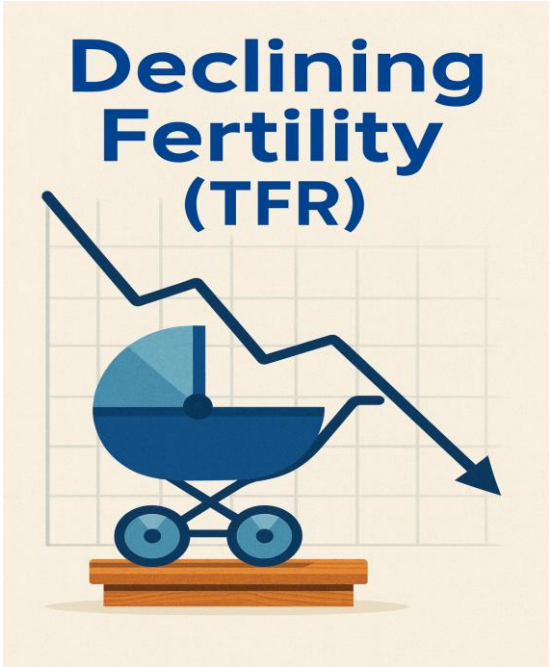
vs

28
World Average

vs

40
Europe Average

Declining Fertility (TFR)



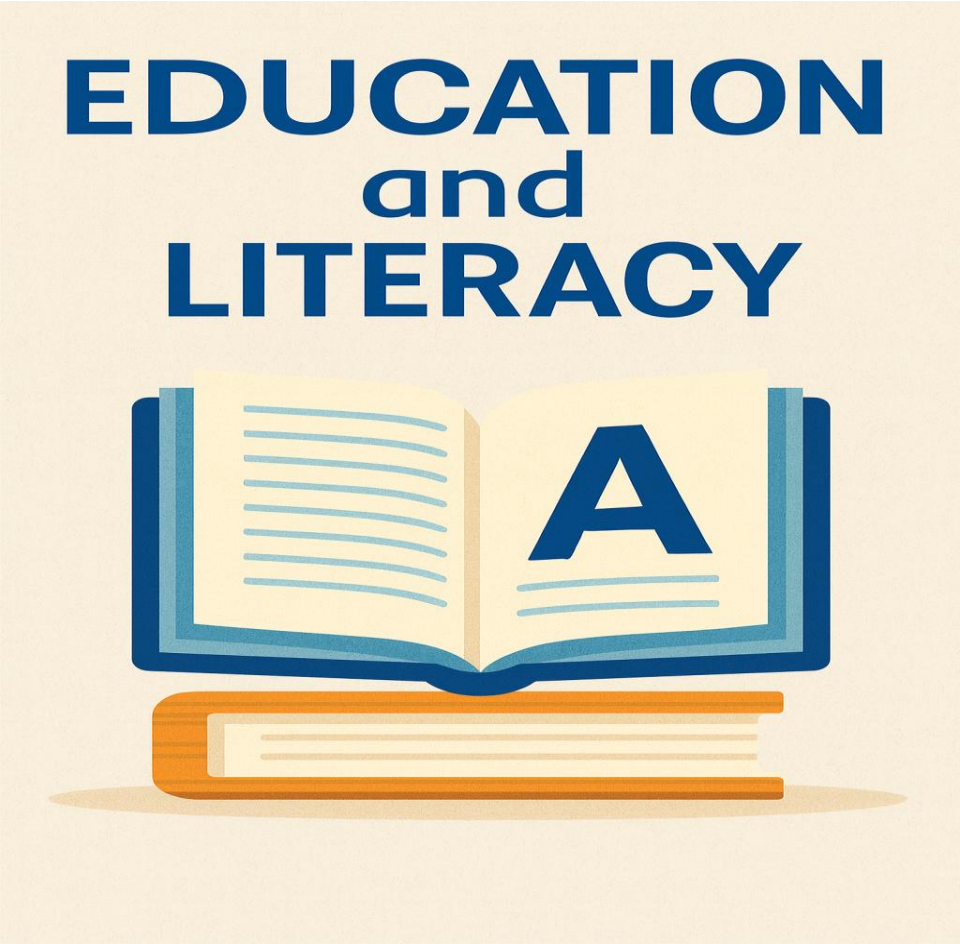


Significant decline in fertility over 15 years, though still higher than replacement level (2.1).

Regional Comparison (2011-13)

Indicator	West Bank	Gaza Strip
Total Fertility Rate (TFR)	3.7	4.5
Crude Birth Rate (2018)	28.0	34.4
Crude Death Rate (2018)	4.0	3.5

Education Status



High Literacy / High Fertility?

Illiteracy (15+ years):

- 1997: 13.9%
- 2017: 3.3%

Female Illiteracy: Dropped from 20.3% to 5.0%.

Usually, high literacy leads to much lower fertility. Palestine is an exception due to political and cultural factors prioritizing large families.

Population Density

West Bank

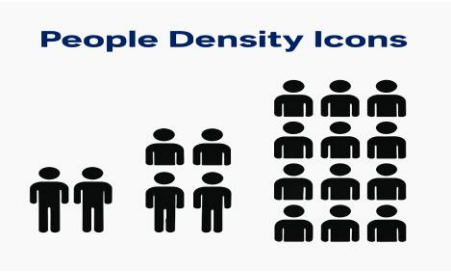
532

Persons / km²

Gaza Strip

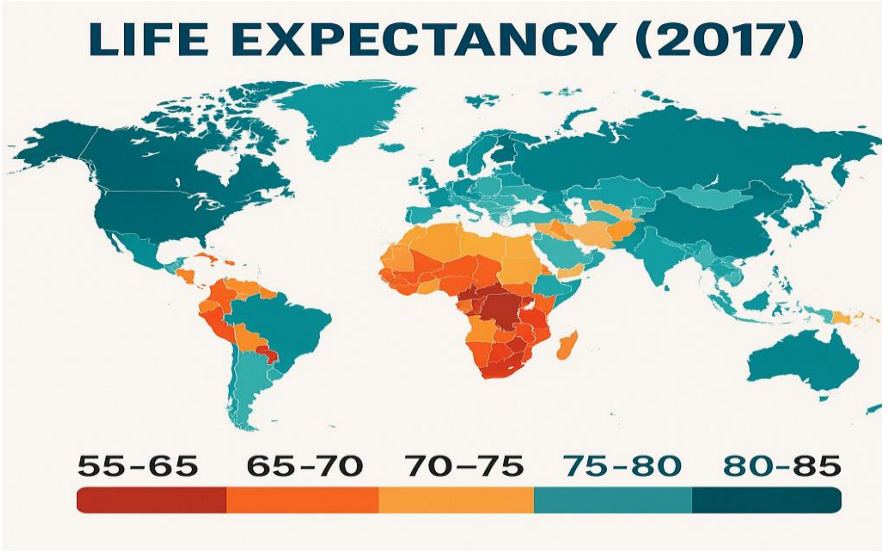
5,324

Persons / km²



Gaza has one of the highest population densities in the world, creating significant public health challenges.

Life Expectancy (2017)



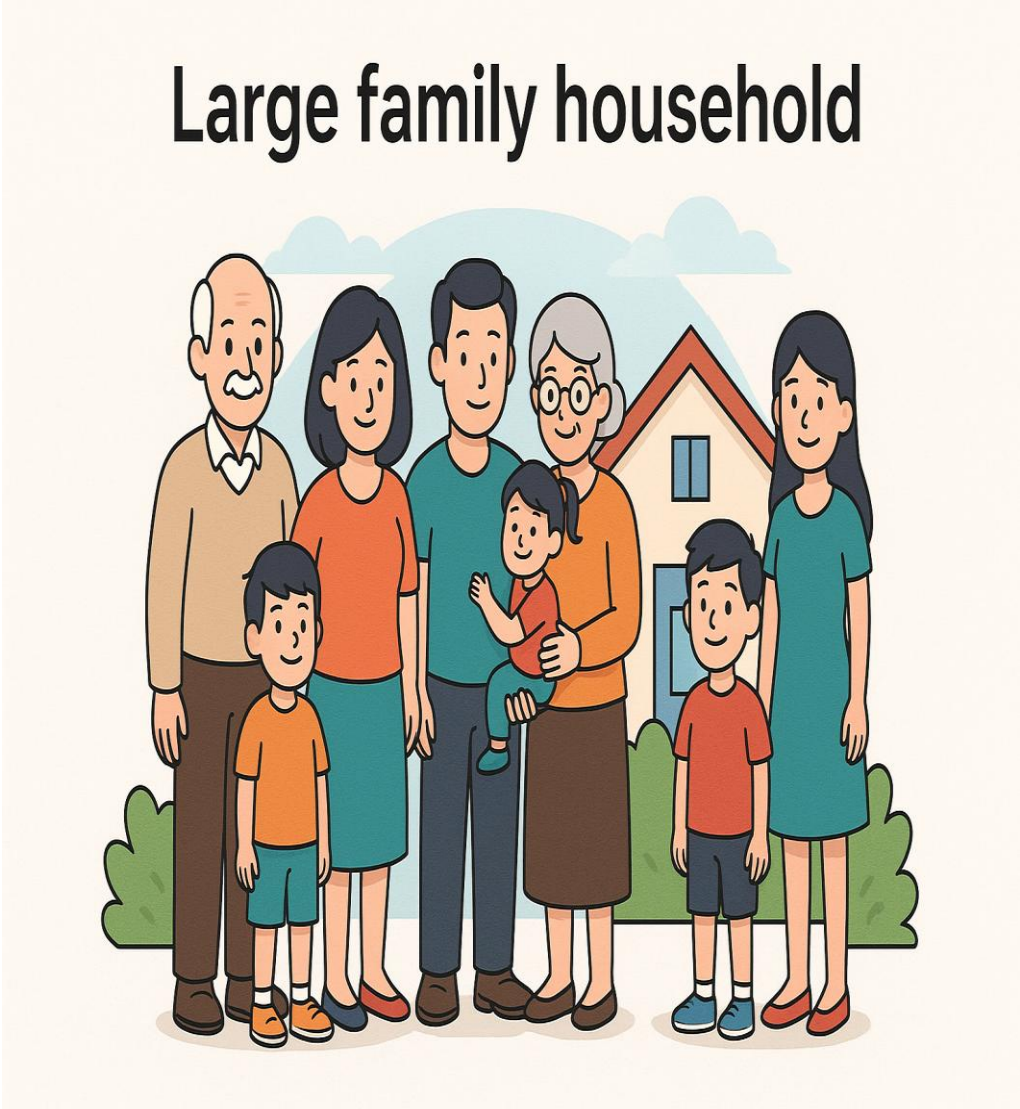


Average Household Size

A Slow Decline

Families are getting smaller, but remain large by international standards.

- 1997: 6.4 persons
- 2012: 5.6 persons
- 2017: 5.1 persons



Demographic Indicators

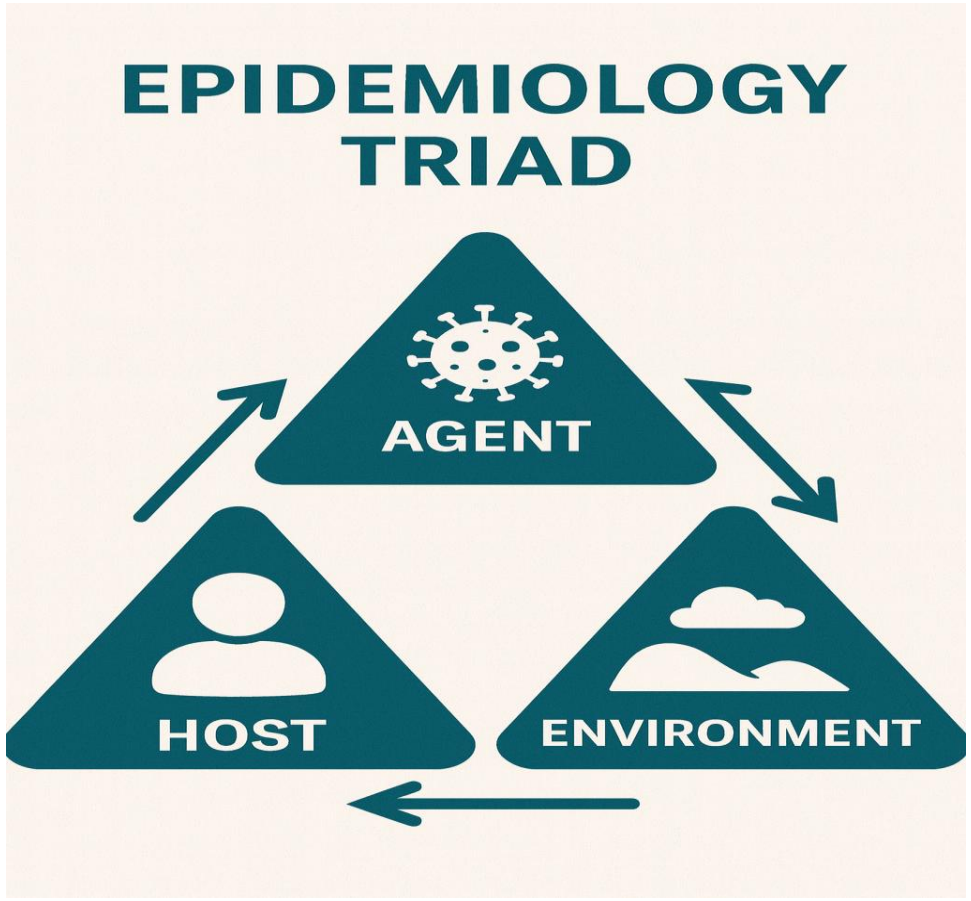
Indicator	2016	2017	2018	2019	2020
Total population**	1,912,267	1,899,291*	1,961,406	2,018,970	2,077,357
% of male**	50.8	50.7	51.0	50.7	50.7
% of female**	49.2	49.3	49.0	49.3	49.3
Male / female ratio (for general pop.)**	103.3	103.3	103.0	102.8	102.7
Proportion of pop. aged under 5 years**	16.7	15.0	14.9	14.9	15.0
Dependency ratio**	82.1	82.1	80.0	79.2	78.6
Median age (years)**	18.4	18.4	19.0	19.2	19.2
Population natural increase rate (%)**	3.3	3.3	3.5	2.9	2.9
Total fertility rate	3.8	3.9	3.7	3.3	3.2
Life expectancy for male**	71.5	71.5	72.0	72.5	72.6
Life expectancy for female**	74.6	74.6	75.0	74.7	74.8
Average family size**	5.7	5.6	5.6	5.5	5.5
No. of new born	58,170	58,954	57,694	55,212	53,687
CBR per 1000 population	30.9	31.0	30.0	27.7	26.2
Percentage of low birth weight (less than 2500gm) of total births	6.3	6.9	7.3	7.3	6.8

*The total number of population was calculated through the census 2017. **Palestinian Central Bureau of Statistics (PCBS).

Applied Epidemiology

Linking Demography to Infectious Disease

Why Demography Matters for Epi



The Epidemiologic Triad

Demography defines the **HOST** (Age, immunity, density).

- **Density** → Increases contact rate.
- **Age** → Determines susceptibility (e.g., childhood diseases).
- **Household Size** → Determines intra-household transmission.

High Density & Transmission

The Gaza Context

With >5,000 people/km², physical distancing is difficult.



High Risk For:

- Airborne diseases (Influenza, COVID-19, TB).
- Contact diseases (Scabies).
- Waterborne diseases (if sanitation fails).

Age Structure & Immunity



Targeting the Young

Since ~40% of the population is under 15 (Expansive Pyramid):

- **Priority:** Routine Immunization (Expanded Program on Immunization) is the #1 defense.
- **Risk Areas:** Schools and Nurseries are amplification sites for outbreaks (Measles, Mumps).
- **Strategy:** School-based health programs are efficient.

Household Transmission



Large Families

Avg size 5.1 means more contacts per home.



Shielding

Protecting the elderly is difficult in multi-generational homes.



Attack Rate

Secondary attack rates for flu/COVID are higher in larger households.

The Dependency Burden

Economic Impact on Health

Dependency Ratio: Ratio of non-workers (children/elderly) to workers.

High youth dependency places a strain on the tax base that funds public health.

Resources are stretched thin between education and pediatric care.



Demographic Momentum Challenges



WASH

Strain on water and sewage infrastructure, increasing cholera/typhoid risk.



Capacity

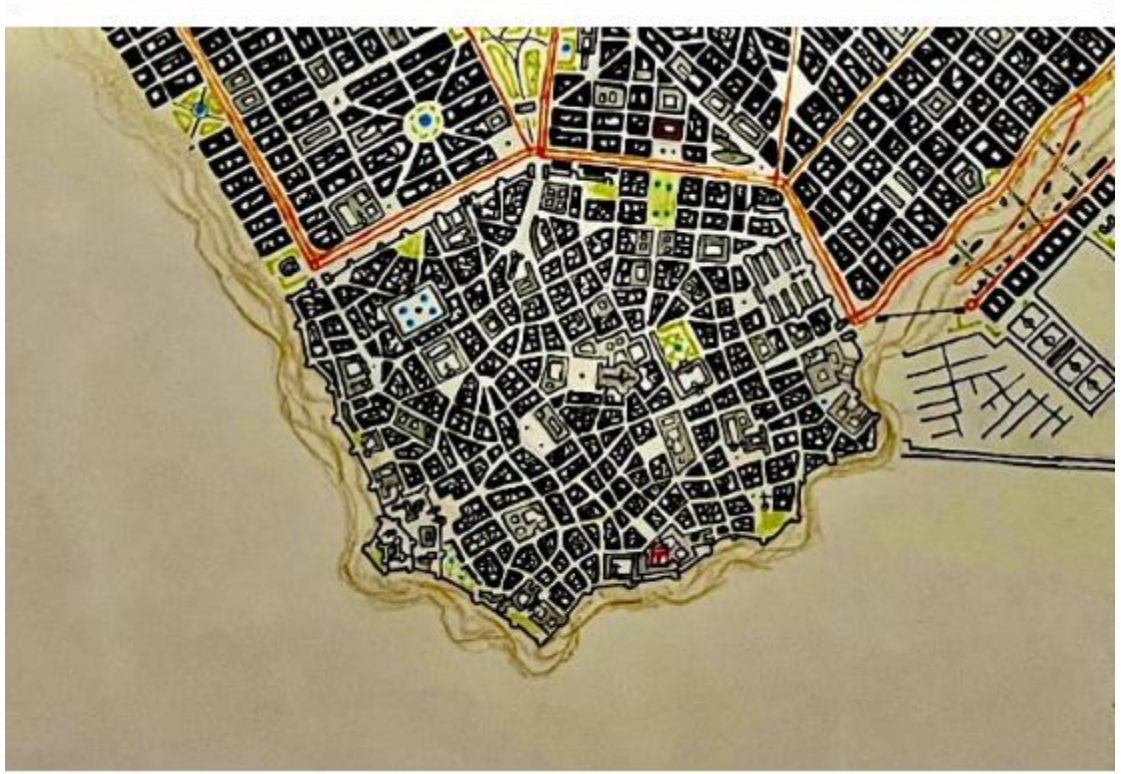
Hospitals constantly overcrowded; bed occupancy rates .



Waste

Increased solid waste generation requires robust disposal to prevent vectors (rats/flies).

Case Study: IPC in Camps



Planning for Density

Infection Prevention & Control (IPC) in Gaza camps requires:

1. **Community Health Workers:** Door-to-door surveillance.
2. **Ventilation:** Improving airflow in schools/clinics.
3. **Triage:** Rapid separation of respiratory cases at clinic entries to protect the waiting crowd.

Summary



Data

Census and Vital Stats are the backbone of health planning.

Transition

Palestine is in Stage 2/3: Lower death rates, slowly declining birth rates.



Risk

High density + Young population = Focus on Immunization and Respiratory IPC.

Assignment

Comparative Analysis

Task: Compare the Age Pyramid of Gaza (2020) with EU (Italy).

- **Scenario:** If a new flu pandemic hits, which age group in each country is most at risk?
- How should Ministry of Health priorities differ?



Thank you for your attentions